Checking Logistic Regression Models

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Slide 1 - notes and edits

Model Selection and Model Fit

On Monday, we compared possible logistic regression models.

- What if the models we considered all miss something?
- How can we find where a model fits poorly?
- Does an unusual observation change our conclusions?

A useful model should describe the data adequately as well as compare favorably with alternatives.

Slide 2 - notes and edits

Deviance

The **saturated model** fits the recorded responses exactly.

$$D_{\text{residual}} = 2\big(\ell_{\text{saturated}} - \ell_{\text{fitted}}\big),$$
$$D_{\text{null}} = 2\big(\ell_{\text{saturated}} - \ell_{\text{intercept-only}}\big).$$

- Smaller residual deviance means a closer fit to the saturated model.
- Deviance decreases as we add terms to a nested model.

How much improvement would justify adding more terms?

Slide 3 - notes and edits

Marijuana Use

Students at one high school near Dayton reported whether they had *ever* used marijuana.

```
> Marijuana <- read.table(file.path(lecture_dir, "data",
+     "Marijuana.dat"), header = TRUE)
> Marijuana
   race gender yes  no
1 white female 420 620
2 white   male 483 579
3 other female  25  55
4 other   male  32  62
```

Each row is a binomial count at one combination of race and gender. "Other" is the dataset's race category.

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An Additive Logistic Model

$$\text{logit}(\pi) = \alpha + \beta_1 \,\text{gender} + \beta_2 \,\text{race}.$$

```
> fit <- glm(cbind(yes, no) ~ gender + race, family = binomial,
+     data = Marijuana)
> round(coef(summary(fit)), 4)
            Estimate Std. Error z value Pr(>|z|)
(Intercept)  -0.8303     0.1685 -4.9266   0.0000
gendermale    0.2026     0.0852  2.3785   0.0174
racewhite     0.4437     0.1677  2.6466   0.0081
```

What restriction does this model impose on the gender odds ratio?

Slide 5 - notes and edits

Goodness of Fit Using Deviance

For g binomial groups and p fitted parameters,

$$G^2 = 2\sum_{\text{cells}} O \log\left(\frac{O}{E}\right), \qquad df = g - p.$$

The cells include both successes and failures in every group; a zero observed count contributes zero.

- H_0: the working model correctly describes the group probabilities.
- Large deviance provides evidence of lack of fit.
- Use a right-tail χ^2_{g-p} probability when fitted counts are sufficiently large.

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Goodness of Fit in R

There are four groups and three model parameters: $df = 4 - 3 = 1$.

```
> c(deviance = deviance(fit), df = df.residual(fit), p_value =
    pchisq(deviance(fit),
+     df.residual(fit), lower.tail = FALSE))
deviance       df  p_value
0.057982 1.000000 0.809715
```

- No evidence that the additive model fits poorly.
- This does not establish that the model is exactly correct.

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Observed and Fitted Counts

$$\widehat{\text{successes}}_i = n_i\hat{\pi}_i, \qquad \widehat{\text{failures}}_i = n_i(1-\hat{\pi}_i).$$

```
> counts <- transform(Marijuana, fitted_yes = fitted(fit) *
+     (yes + no), fitted_no = (1 - fitted(fit)) * (yes + no))
> counts[c("race", "gender", "yes", "fitted_yes", "no",
    "fitted_no")]
   race gender yes fitted_yes  no fitted_no
1 white female 420    420.714 620   619.286
2 white   male 483    482.286 579   579.714
3 other female  25     24.286  55    55.714
4 other   male  32     32.714  62    61.286
```

All eight fitted counts exceed about 5, supporting the chi-square approximation here.

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The Omitted Interaction

With two binary predictors, adding the interaction saturates the four-group model.

```
> fit_int <- update(fit, . ~ . + gender:race)
> anova(fit, fit_int, test = "LRT")
Analysis of Deviance Table
Model 1: cbind(yes, no) ~ gender + race
Model 2: cbind(yes, no) ~ gender + race + gender:race
  Resid. Df Resid. Dev Df Deviance Pr(>Chi)
1         1      0.058
2         0      0.000  1    0.058     0.81
```

This tests whether the gender odds ratio varies by race. It does **not** test whether the main effects are zero.

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Grouped and Ungrouped Data

The same subjects can be recorded in two ways.

	Grouped	Individual
Rows	4 groups	2,276 people
Response	Yes/no counts	0 or 1
Trials per row	n_i	1
Fitted probabilities	$\hat{\pi}_i$	Same within each group

Do you expect the coefficients to change? What about the residual deviance?

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The Same Estimated Effects

After expanding the counts into individual responses:

```
> fit_people <- glm(y ~ gender + race, family = binomial,
+     data = people)
> round(cbind(grouped = coef(fit), individual =
    coef(fit_people)),
+     4)
            grouped individual
(Intercept) -0.8303    -0.8303
gendermale   0.2026     0.2026
racewhite    0.4437     0.4437
```

The response format changes; the additive model still assigns one probability to each race–gender combination.

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Different Residual Deviances

```
> rbind(grouped = c(deviance = deviance(fit), df =
    df.residual(fit)),
+     individual = c(deviance = deviance(fit_people), df =
    df.residual(fit_people)))
             deviance   df
grouped       0.057982    1
individual 3086.598201 2273
```

- Grouped saturation: one probability for each of four groups.
- Individual saturation: one probability for each person's outcome.

The fitted model is the same, but the saturated benchmark changes.

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What Can the Saturated Model Fit?

Consider two people with identical predictors: one says yes, the other no.

- Grouped saturated model: $\hat{\pi} = 1/2$ for the pair.
- Individual saturated model: $\hat{\pi}_1 = 1$ and $\hat{\pi}_2 = 0$.

On the individual-data likelihood scale,

$$L_{\text{common } p} = p(1-p) \leq \tfrac{1}{4}, \qquad L_{\text{individual saturation}} = 1.$$

The individual saturated model can fit which person said yes, rather than just how many said yes. Binomial combinatorial factors cancel in each deviance ratio.

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Nested Model Comparisons Still Agree

Subtracting deviances cancels the saturated-model benchmark:

$$D_{\text{smaller}} - D_{\text{larger}} = 2(\ell_{\text{larger}} - \ell_{\text{smaller}}).$$

```
> fit_people_int <- update(fit_people, . ~ . + gender:race)
> c(grouped = deviance(fit) - deviance(fit_int), individual =
    deviance(fit_people) -
+     deviance(fit_people_int))
   grouped individual
  0.057982   0.057982
```

The interaction test agrees in either representation. The usual likelihood-ratio theory applies under its regularity conditions.

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Continuous Predictors and Individual Responses

For individual Bernoulli data, do **not** automatically compare residual deviance with residual degrees of freedom using a chi-square distribution.

- Each recorded response is only 0 or 1.
- More people do not produce large binomial counts at each distinct predictor setting.
- Compare specific larger models to investigate possible lack of fit: a quadratic term, an interaction, or another relevant predictor.

Judge the practical change in predictions as well as the test result.

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Graduate Admissions at University of Florida

Model admission probability by department, with no gender effect:

$$\text{logit}(\pi_{ik}) = \alpha + \beta_k.$$

```
> Admissions <- read.table(file.path(lecture_dir, "data",
+     "Admissions.dat"), header = TRUE)
> Admissions$gender <- factor(Admissions$gender, levels = c(0,
+     1), labels = c("male", "female"))
> adm_dept <- glm(cbind(yes, no) ~ department, family =
    binomial,
+     data = Admissions)
> c(deviance = deviance(adm_dept), df = df.residual(adm_dept),
+     p_value = pchisq(deviance(adm_dept),
    df.residual(adm_dept), lower.tail = FALSE))
  deviance         df    p_value
44.7351647 23.0000000  0.0042823
```


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Residuals Locate Lack of Fit

For group i, the observed and fitted success counts are y_i and $n_i\hat{\pi}_i$.

$$\text{Standardized residual}_i = \frac{y_i - n_i\hat{\pi}_i}{\widehat{\text{SE}}(Y_i - n_i\hat{\pi}_i)}.$$

- Positive: more successes than predicted.
- Negative: fewer successes than predicted.
- Magnitudes around 2 or 3 suggest groups worth examining.

These are investigation prompts: sparse counts and examining many groups complicate their interpretation.

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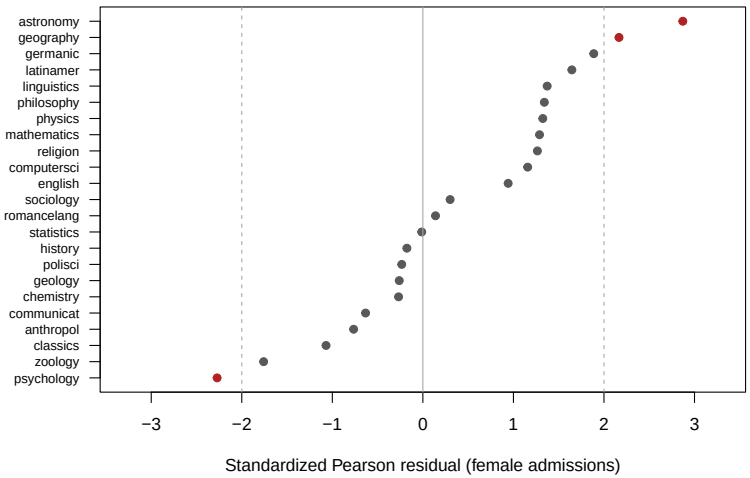
Admissions Residuals in R

```
> Admissions$std_res <- rstandard(adm_dept, type = "pearson")
> subset(Admissions, gender == "female" & abs(std_res) >
+     2, select = c(department, yes, no, std_res))
   department yes  no std_res
4   astronomy   6   0  2.8710
16  geography   9   1  2.1664
36 psychology   2 123 -2.2722
```

These are residuals for female admissions under the department-only model. Which departments differ, and in which direction?

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Admissions Residuals by Department



The model misses different departments in different directions.

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Adding a Common Gender Effect

```
> adm_gender <- update(adm_dept, . ~ . + gender)
> anova(adm_dept, adm_gender, test = "LRT")
Analysis of Deviance Table
Model 1: cbind(yes, no) ~ department
Model 2: cbind(yes, no) ~ department + gender
  Resid. Df Resid. Dev Df Deviance Pr(>Chi)
1        23       44.7
2        22       42.4  1     2.38     0.12
```

A common gender effect provides little improvement. It cannot describe effects that vary across departments. Some admission counts are sparse, so treat the global chi-square goodness-of-fit approximation cautiously. These comparisons alone do not establish a cause for the differences.

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Residuals and Influence

- A **residual** describes a discrepancy between an observed and fitted response.
- **Influence** describes how much an observation changes fitted coefficients or predictions.

Refitting without a flagged group can help assess sensitivity. An unusual group may have little influence. A flag is a reason to investigate, rather than an instruction to remove data.

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Heart Disease and Blood Pressure

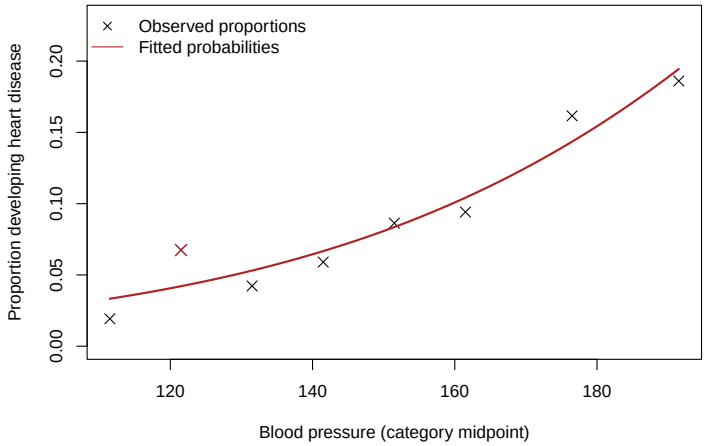
Eight blood-pressure groups; response is heart disease during six years of follow-up among men aged 40–59.

```
> HeartBP <- read.table(file.path(lecture_dir, "data",
    "HeartBP.dat"),
+     header = TRUE)
> heart_fit <- glm(cbind(y, n - y) ~ bp, family = binomial,
+     data = HeartBP)
> c(deviance = deviance(heart_fit), df =
    df.residual(heart_fit),
+     p_value = pchisq(deviance(heart_fit),
    df.residual(heart_fit),
+         lower.tail = FALSE))
deviance       df  p_value
 5.90916  6.00000  0.43344
```

These are grouped binomial counts, despite using numerical blood-pressure scores. Check the fitted counts before relying on the approximation.

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Observed Proportions and Fitted Probabilities



The second group stands out, while the overall pattern is reasonably described by the fitted curve.

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Sensitivity to One Group

```
> heart_without2 <- update(heart_fit, data = HeartBP[-2,
+     ])
> rbind(all_groups = coef(heart_fit), without_group2 =
    coef(heart_without2))
              (Intercept)       bp
all_groups        -6.0820 0.024338
without_group2    -7.0841 0.030309
```

- The estimated blood-pressure slope increases when group 2 is omitted.
- The fitted association remains positive in both fits.
- This removes an entire binomial group, not one person.

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Review

1. What does the marijuana residual-deviance test tell us about the interaction?
2. Why does expanding counts into individual responses change residual deviance?
3. Why might adding one gender coefficient fail to address the admissions lack of fit?
4. How does a large residual differ from an influential observation?

What would you examine next if your fitted model looked inadequate?

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Appendix: Pearson and Standardized Residuals

The estimated binomial standard deviation is

$$\sqrt{n_i\hat{\pi}_i(1-\hat{\pi}_i)}.$$

Dividing the raw count residual by this gives the **Pearson residual**. The **standardized Pearson residual** also adjusts for fitting:

$$r_i^* = \frac{y_i - n_i\hat{\pi}_i}{\sqrt{n_i\hat{\pi}_i(1-\hat{\pi}_i)(1-h_{ii})}}.$$

Leverage h_{ii} accounts for how strongly group i helps determine its own fitted value. The variance expression is a model-based approximation.

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Appendix: Fitted Values and Leverage

```
> data.frame(probability = fitted(fit), leverage =
    hatvalues(fit),
+      pearson = residuals(fit, type = "pearson"), standardized
    = rstandard(fit,
+          type = "pearson"))
  probability leverage   pearson standardized
1     0.40453  0.96492 -0.045129     -0.24096
2     0.45413  0.96662  0.044023      0.24096
3     0.30357  0.48045  0.173685      0.24096
4     0.34802  0.58801 -0.154665     -0.24096
```

- `fitted(fit)`: estimated success probabilities.
- `hatvalues(fit)`: diagonal leverages.
- `rstandard(fit, type = "pearson")`: standardized Pearson residuals.

The four standardized residuals have the same absolute value here: only one residual degree of freedom remains.

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Appendix: Influence Diagnostics in R

```
> heart_diagnostics <- data.frame(bp = HeartBP$bp, std_res =
    rstandard(heart_fit,
+      type = "pearson"), cook = cooks.distance(heart_fit),
    dfbetas_bp = dfbetas(heart_fit)[,
+      "bp"])
> round(heart_diagnostics, 3)
     bp std_res  cook dfbetas_bp
1 111.5  -1.106 0.168       0.492
2 121.5   2.375 1.132      -1.143
3 131.5  -0.945 0.157       0.328
4 141.5  -0.573 0.046       0.082
5 151.5   0.126 0.001       0.008
6 161.5  -0.326 0.008      -0.065
7 176.5   0.652 0.130       0.401
8 191.5  -0.177 0.010      -0.124
```

Cook's distance summarizes influence; DFBETAS describe standardized changes in individual coefficients. R uses an approximation to deletion effects for GLMs. Group 2 has the largest magnitudes here.

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Sources

- Agresti, *An Introduction to Categorical Data Analysis*, third edition, Section 5.2, pp. 130–136.
- Marijuana use: Section 5.2.2; graduate admissions: Table 5.2; heart disease: Table 5.3 and Figure 5.1.
- Data: Alan Agresti's public data repository.

All displayed R results and plots are generated by the companion `model_checking.R` script using the original data files.

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